DS 3003 /ST 3083 - Machine Learning 1
Final Project - Proposal

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Title of the Project: Arrhythmia Classification from Two-Lead ECG Signals

Introduction:

Cardiovascular diseases (CVDs) are the leading cause of death worldwide, accounting for more fatalities annually than any other condition, according to the WHO. Many CVDs develop due to long-term cardiac arrhythmias, which may initially appear harmless but can lead to severe complications, including heart failure. Advances in machine learning and digital signal processing have enabled automated ECG analysis. However, existing models often rely on single-lead ECG data or lack consistency, reducing their reliability and clinical applicability. This project aims to address these limitations by developing a robust classification model based on two-lead ECG signals.

Description of the Problem:

Early detection of arrhythmias is crucial for reducing the risk of CVDs. Long term ECG monitoring is the most effective method, but it is often inaccessible to individuals in developing regions due to high costs and limited medical infrastructure. Our primary objective is to develop a model that accurately classifies different types of cardiac arrhythmias using two-lead ECG signals. The secondary objective is to explore the potential for a cost-effective application that provides a reliable estimate of heart health using minimal lead signals. By utilizing two-lead ECG data and well-extracted features, our project aims to enhance both the accuracy and usability of detecting normal and abnormal heartbeats in patients without the need of long-term monitoring. Furthermore, an automated and reliable classification system can aid remote monitoring, improving early detection and intervention in resource-limited settings.

Description of the Dataset:

The dataset sourced from Kaggle (originally from **MIT-BIH Arrhythmia Database**) contains **100,689 observations** and **34 variables** extracted from ECG signals. Each entry includes a **record** identifier (patient ID) and a **type** label, classifying heartbeats into five categories: **Normal (N), Supraventricular ectopic beat (S), Ventricular ectopic beat (V), Fusion beat (F), and Unknown beat (Q)**. The dataset features data from two ECG leads (Lead II and Lead V), with each lead contributing 16 extracted features. These features include **RR intervals** (pre-RR and post-RR), **heartbeat intervals** (PQ Interval, QT Interval, ST Interval, QRS Duration), **amplitude features** (P peak, T peak, R peak, S peak, Q peak), and **morphology features** (five QRS morphology components per lead). All 32 extracted features are numerical features and the type is the only categorical feature exist.

Comments:

The dataset description on Kaggle suggests the inclusion of an "Average RR Interval" feature; however, it is not present in the dataset. Since this dataset is specifically designed for heartbeat classification, our primary objective remains focused on arrhythmia detection. However, our secondary objective, exploring a potential application for cost-effective and accessible ECG monitoring could lead to new opportunities for real-world deployment, particularly benefiting patients with limited access to medical resources.

References:

Link to the dataset - https://www.kaggle.com/datasets/sadmansakib7/ecg-arrhythmia-classification-dataset?select=MIT-BIH+Arrhythmia+Database.csv

Sakib, S., Fouda, M. M., & Fadlullah, Z. M. (2021). Harnessing artificial intelligence for secure ECG analytics at the edge for cardiac arrhythmia classification. In *CRC Press eBooks* (pp. 137–153). https://doi.org/10.1201/9781003028635-11